JobTrack

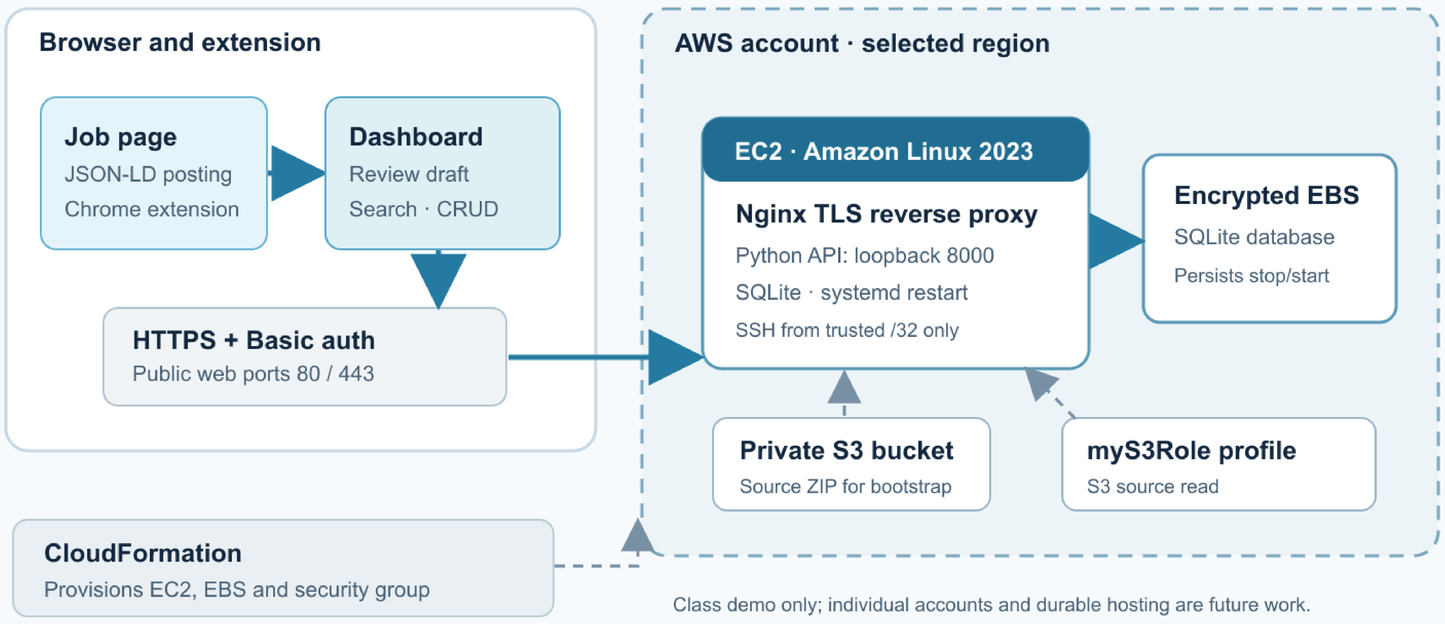
A cloud-hosted job application tracker

CSE 4333 Cloud Computing · Group 42

Ahmad Salem · NoorAldeen Waleed

JobTrack public HTTPS architecture

User-reviewed capture · shared demo password · public HTTPS



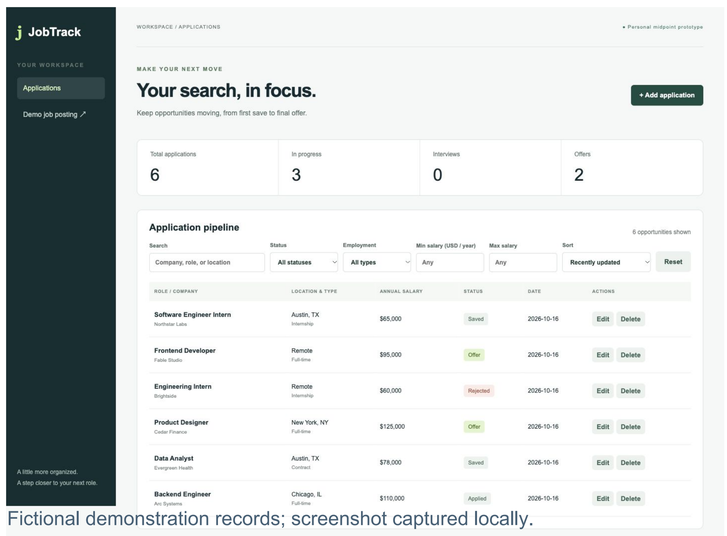
Project summary

JobTrack keeps a student's job applications in one searchable pipeline. It stores company, role, location, annual salary, status, date, link, and notes.

The dashboard, Python API, and SQLite support tracking and job discovery. Remotive works without a key; Indeed and LinkedIn use optional JSearch. A Chrome extension reads structured postings.

Verified public HTTPS AWS sandbox demo: Nginx with a shared password proxies to EC2's loopback app. CloudFormation provisions EC2 and encrypted EBS; private S3 supplies source.

Application dashboard



Fictional demonstration records; screenshot captured locally.

Introduction · users

Students and recent graduates often manage many applications at once and lose track of dates and next steps.

JobTrack keeps opportunities, status changes, salary, and notes together so a user can see their pipeline and act on it.

User workflow

Find a listing or import a posting draft; review fields; save; update status; search and filter; delete old entries.

Remotive and optional JSearch provide job discovery. The extension reads Schema.org JobPosting JSON-LD. JobTrack does not apply to employers.

Conclusion

JobTrack puts discovered opportunities and application progress in one searchable place. It is a working password-protected class demonstration.

AWS deployment and performance use measured results. Per-user accounts and durable public hosting remain future work.

Application logic

The responsive dashboard calls same-origin JSON endpoints. The Python API validates fields and stores UTC timestamps in SQLite.

Endpoints cover health, application CRUD, /api/jobs discovery, /api/job-sources status, and saving a listing. Search and filters combine; sort keys are allowlisted.

Cloud services

CloudFormation creates EC2, encrypted EBS, and public web rules. SSH is /32-limited; myS3Role reads the private S3 source ZIP.

Nginx serves HTTPS with a shared demo password, then proxies to the loopback Python app.

Alternatives considered

Per-user accounts would isolate applicants. RDS or DynamoDB would improve managed storage; serverless hosting could scale differently.

The one-instance SQLite design is appropriate for a small class demo, but it has no high availability or automatic backup.

Challenges and lessons

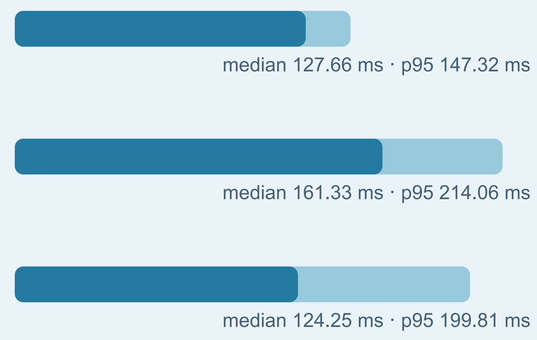
Browser QA exposed a Reset-button conflict. Final API review exposed unclosed SQLite connections. Both were fixed and regression-tested.

Public access required TLS and a password at the reverse proxy. The sandbox session is time-limited; the EBS root volume needs backup before termination.

Team contributions

Ahmad Salem and NoorAldeen Waleed — equal co-developers. NoorAldeen built the core tracker and job-discovery features. Ahmad handled final cloud deployment, testing, documentation, poster, and slides. Both shaped the project design.

Performance evaluation



50 sequential GET requests per endpoint after 5 warmups; 5 fictional records. t3.micro, us-east-1; includes the client-to-EC2 SSH tunnel. Small single-user sample.

Verification

23 Python tests and 5 extractor tests pass. The browser checked draft review, save, edit, refresh, filters, sorting, deletion, and job discovery.

Cloud CRUD, restart persistence, and Remotive discovery passed. Public HTTPS returned 401 without a password and 200 with one. The installed extension toolbar was not tested.

Machine learning

Not applicable. JobTrack uses rules and a relational database; it has no trained model.

Future work

Add individual accounts and user isolation before real applicant data. Move to managed backup/database, add monitoring, and broaden posting extraction.

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